

Ice-ocean Antarctica focus group – draft protocol for ice sheet modellers

1. Overall approach

Sub-shelf melt rates in ISMIP7 are, similar to ISMIP6, modelled using melt rate parameterisations. Since ISMIP6, a number of melt parameterisations have been developed, evaluated and inter-compared. So far, none of them is unanimously accepted as the best option. Every group is therefore free to make their own choice on which parameterisation of sub-shelf melting they want to use, based on physics or machine learning, as long as it can be driven by ocean properties from the CMIP models. We also leave it to the groups to make their own choice on melting in the grounding zone or melting at vertical ice faces (the terminus after ice shelf collapse or the calving front if an ice shelf is present). We summarise the strategy for ISMIP7 in Figure 1.

To force the melt parameterisation, we provide input ocean conditions around Antarctica, bias-corrected and extrapolated from CMIP6/7 models with the test case being the historical and extended SSP5-8.5 scenarios of the CESM2-WACCM model, from 1850 to 2299. We also provide an updated ocean climatology for the present-day state which is used for bias correction.

Moreover, a novelty compared to ISMIP6 is that we provide datasets and a protocol to calibrate the parameters of the melt parameterizations from consistent data coming from both observations and ocean model projections. More observations and ocean–ice-shelf projections have been produced since the design of the ISMIP6 protocol, and can now be used to better constrain the melt sensitivity to ocean warming in ISMIP7. We believe that the melt parameterisation should not be used as a simple tuning parameter for the ice sheet mass loss, it should be tuned to provide plausible melt rates and melt rate sensitivities to oceanic changes.

We propose a MeltMIP to evaluate and inter-compare the melt rates produced by the ISMIP7 ice sheet models and parameterisations (to be done later on).

All datasets can be obtained from the ISMIP7 Forcing Globus endpoint /ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers. Note that those are work in progress, hence please do use with care, let us know of any issues, and get in touch with us

if you want to use the data for any other projects until they are finalised and published. If any questions arise, please get in touch!

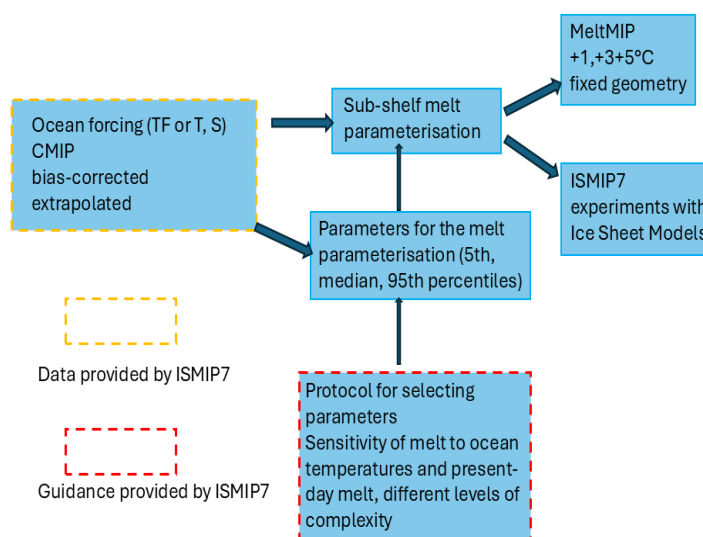


Figure 1: Antarctic sub-shelf melt-rate approach flow chart. Extrapolated and bias-corrected ocean forcing fields are provided to the ice sheet modellers by the ISMIP7 AIS ocean forcing working group. A protocol for selecting parameters in the melt-rate parameterisation of modeller's choice is provided. Each ice sheet model must tune their melt parameterisation parameters following this protocol to calculate sub-shelf melt for the

projections. The parameterisation's reaction to thermal forcing changes will be tested in a simple MeltMIP.

2. ISMIP

2.1 Melt parameterisations

In contrast to ISMIP6, modellers are open to select their own melt parameterisation. If modellers would like to use the quadratic-dependency melt parameterisation that was proposed in ISMIP6 (Jourdain et al., 2020), please incorporate the recommended updates. However, we ask modellers to follow the calibration protocol provided below to calibrate their melt rate parameterisation as it is coded in their model and at the grid resolution they use.

Recommended updates to the quadratic melt parameterisation as developed for ISMIP6 (Nowicki et al., 2020; Jourdain et al., 2020, ISMIP6-Projections-Antarctica, and Favier et al. 2019) with the revised formulation from Burgard et al. (2022) based on Jenkins et al. (2018):

$$m = K \cdot \sin\theta \cdot \frac{\rho_{oc}}{\rho_{ice}} \left(\frac{c_{oc}}{L_{ice}} \right)^2 \beta_S S_{loc} \frac{g}{2|f|} |T_{loc} - T_{f,loc}| (T_{loc} - T_{f,loc}) \quad (1)$$

where K is the tuning parameter. $\sin\theta$ is the slope of the ice draft, which can be either assumed constant (average over all ice-draft slopes under Antarctic ice-shelf points) or locally dependent on the simulated ice-sheet topography. Since Favier et al. (2019), there has not been any further evidence that the “semi-local” (referred to as “non-local” in Jourdain et al., 2020) version of the quadratic parameterisation gives better results than the “local” version (Burgard et al., 2022), so any of these versions is welcome in ISMIP7.

As in ISMIP6, adding a temperature correction δT_b in individual basins may help obtain a good match between the present-day ocean temperatures and melt rates. These temperature corrections should maintain the present-day thermal forcing in a plausible range, i.e. not significantly below 0°C or above 5°C.

2.2 Datasets for melt parameterisations

We here list all datasets we provide for running the ISMIP experiments.

2.2.1. Ocean forcing for ISMIP projections

Ocean forcing for the projections stems from CMIP simulations, with the test case being CESM2-WACCM between 1850 and 2300.

In a similar way as in ISMIP6, we directly provide bias-corrected ocean data extrapolated to any possible location of a simulated ice shelf draft on a polar stereographic grid. Our workflow to process the CMIP ocean outputs in ISMIP7 differs from ISMIP6 in the details and is as follows:

1. Convert the CMIP monthly potential temperature and practical salinity to conservative temperature (CT) and absolute salinity (SA) that better conserve energy in the mixing of water masses during the interpolation/extrapolation/averaging process (McDougall 2025).
2. Remap to the 8km x 8km x 60m standard stereographic grid used in ISMIP7 using bilinear interpolation in the horizontal and linear interpolation in the vertical.

3. Extrapolate to all the grid points that were not covered by ocean points in the CMIP model following 4 steps (the 16 IMBIE basins were first extrapolated to the ocean domain as in ISMIP6):
 - a. Horizontal extrapolation in a given IMBIE basin, only towards adjacent ocean points (open ocean and present-day ice-shelf cavities from Bedmap3).
 - b. Horizontal interpolation beyond present-day ice shelves, going across the ice but not across bathymetric ridges.
 - c. Horizontal extrapolation everywhere at level (k=2), not accounting for IMBIE basins, even if the bedrock is above sea level, in order to extrapolate downward everywhere later on (level k=1 corresponds to the surface mixed layer and therefore has specific properties).
 - d. Vertical extrapolation from above for all the remaining points (and from below for k=1).
4. Bias correction following the ISMIP6 method (Jourdain et al., 2020), i.e. adjustment of the mean state in the geographical space, by removing the 1995–2014 model climatology and adding the new climatology of Zhou et al. (2025, see next subsection). See below for further justification of the bias correction method.
5. Compute the thermal forcing (TF) using the linearisation of the freezing temperature of Jourdain et al. (2017) for CT and SA.
6. Compute annual means (CT, SA and TF) [option to save the RMS TF instead of the mean TF, to be discussed later].
7. Convert to potential temperature (thetao) and practical salinity (so) for parameterisations involving the calculation of a density based on these variables.

All the framework is provided in <https://github.com/ismip/ismip7-antarctic-ocean-forcing.git>. The extrapolation steps are similar to the one that were used in the PROTECT project (e.g., Caillet et al., 2025). Some investigations towards other bias correction methods were done, in particular a correction method in the (T,S) space (Naughten et al., 2023) and a correction method for the T and S cumulative distribution functions (Verjans et al., 2023), but the results were not convincing [TO BE DEVELOPED IN PROTOCOL PAPER]. Note that the dataset can be used directly by models without computing anomalies or selecting reference observations. If modeling groups need to compute anomalies, they should be computed with respect to the same 1995-2014 period used for the bias correction.

The thetuo, so and TF fields are provided in the following files on Globus:

CESM2-WACCM Historical

directory:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/biascorr_cmip/CESM2-WACCM/historical/zhou_annual_30_sep/Oyr/thetao_so_tf

files:

so_Oyr_CESM2-WACCM_historical_r1i1p1f1_ismip8km_60m_185001-201412.nc
 tf_Oyr_CESM2-WACCM_historical_r1i1p1f1_ismip8km_60m_185001-201412.nc
 thetuo_Oyr_CESM2-WACCM_historical_r1i1p1f1_ismip8km_60m_185001-201412.nc

CESM2-WACCM SSP5-8.5

directory:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/biascorr_cmip/CESM2-WACCM/ssp585/zhou_annual_30_sep/Oyr/thetao_so_tf

files:

so_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_201501-210012.nc
so_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_210101-215012.nc
so_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_215101-220012.nc
so_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_220101-225012.nc
so_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_225101-229912.nc
tf_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_201501-210012.nc
tf_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_210101-215012.nc
tf_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_215101-220012.nc
tf_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_220101-225012.nc
tf_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_225101-229912.nc
thetao_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_201501-210012.nc
thetao_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_210101-215012.nc
thetao_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_215101-220012.nc
thetao_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_220101-225012.nc
thetao_Oyr_CESM2-WACCM_ssp585_r1i1p1f1_ismip8km_60m_225101-229912.nc

2.2.2 Ocean forcing for melt parameterisation calibration: present-day ocean climatology

For the calibration of the melt parameterisation, we provide an ocean present-day climatological state for the Southern Ocean, that is based on the dataset produced for the OCEAN:ICE project:

Shenjie Zhou, Pierre Dutrieux and Claudia Giulivi (2024). Southern Ocean (90°S-45°S) conservative temperature and absolute salinity profiles compilation (OCEAN:ICE D1.1). <https://doi.org/10.17882/99787>

A more recent version will be available here: <https://doi.org/10.17882/103946>. This dataset represents observations between 1st January 1972 and July 2024. For ISMIP, it has been remapped, gaps have been filled, and the dataset has been extrapolated into ice shelf cavities and currently grounded ice, following the same method as described above for the CMIP models. [Comparison to the new climatology of Yamazaki et al. (2025) is planned].

Extrapolated thetao and so fields are provided in the following files on Globus:

directory:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/climatology/zhou_annual_30_sep

files:

OI_Climatology_ismip8km_60m_so_extrap.nc
OI_Climatology_ismip8km_60m_tf_extrap.nc
OI_Climatology_ismip8km_60m_thetao_extrap.nc

2.2.3 CMIP climatology and bias

Some modeling groups may wish to make use of the CMIP model climatologies or the bias fields (e.g. for working with anomalies). However, modelers will need to bear in mind that, since these fields were computed in terms of conservative temperature (ct) and absolute salinity (sa), we provide them in this form and do not attempt to convert them to potential temperature (thetao) and practical salinity (so).

CESM2-WACCM Climatology

directory:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/biascorr_cmip/CESM2-WACCM/climatology

files:

ct_CESM2-WACCM_climatology_r1i1p1f1_ismip8km_60m_199501-201412.nc
sa_CESM2-WACCM_climatology_r1i1p1f1_ismip8km_60m_199501-201412.nc

CESM2-WACCM Bias

directory:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/biascorr_cmip/CESM2-WACCM/bias/zhou_annual_30_sep/

files:

ct_CESM2-WACCM_bias_r1i1p1f1_ismip8km_60m_199501-201412.nc
sa_CESM2-WACCM_bias_r1i1p1f1_ismip8km_60m_199501-201412.nc

2.2.4 Melt rate dataset for melt parameterisation calibration: present-day

For calibration, modelled melt rates will be compared to observed melt rates. We here provide the melt rate dataset with average melt rates between 1992 and 2017 by Paolo et al., 2023 (<https://nsidc.org/data/nsidc-0792/versions/1>), (bilinearly) remapped to the ISMIP 8km grid and aggregated over the imbie2 basins as used in ISMIP. To capture a wider range of plausible melt rates, error estimates are based on combined errors by Paolo et al., 2023, and Adusumilli et al., 2020.

Melt rate fields and imbie-basin aggregated values are provided as:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/meltmip/melt_paolo_err_adusumilli_ismip8km.nc

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/meltmip/Melt_Paolo_Err_Adusumilli_imbie2.csv

2.2.5 Ocean forcing & melt rates for melt parameterisation calibration: ocean modelling data

For the calibration of melt parameterisations, we provide ocean forcing data (thermal forcing as well as potential temperature and practical salinity) from ocean model simulations that simulate large-scale changes in ice shelf cavities with the aim to calibrate melt rate parameters, with a procedure detailed below. Data is from simulations by Mathiot et al. 2023 for all Antarctic basins; Timmermann & Goeller 2017 for the Filchner-Ronne Ice Shelf. Further data will be incorporated, planned at the moment are simulations by Pelle et al.,

2021 for Totten, Naughten et al., 2021 for Filchner-Ronne, Naughten et al., 2018, Antarctic-wide, Jourdain et al. 2022 and Naughten et al., 2023 for the Amundsen Sea.

For each ocean model simulation, we provide a “cold” and a “warm” ocean state (thermal forcing, potential temperature, practical salinity) on the ISMIP ocean grid, ready to apply in the melt parameterisations. These states are obtained as averages over 20-year periods of the current and the flipped-cavity conditions, usually at the end of the simulations. Alongside ocean forcing, we provide the modelled melt rates on the standard ISMIP 8km grid, together with an uncertainty estimate based on the variability of melt rates during the averaging period.

directory:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/meltmip/Ocean_Modelling_Data/

files:

Mathiot23_cold_T/S/TF/m.nc

Mathiot23_warm_T/S/TF/m.nc

TimmermannUndGoeller2017_cold_T/S/TF/m.nc

TimmermannUndGoeller2017_warm_T/S/TF/m.nc

2.2.6 Ocean forcing & melt rates for melt parameterisation calibration: ocean observations [tbd]

We further plan to incorporate observational data of bulk melting (with error estimates) and ocean conditions for Pine Island (Dutrieux et al., 2014) and Dotson ice shelves (Jenkins et al., 2018) that span a range of thermal forcing and melt regimes at the interannual time scales.

[tbd]

2.2.7 Basins

Basins are aggregated from the IMBIE2 dataset as described in Jourdain et al. (2020). Where multiple basins cross through major ice shelves (Ross, and Filchner-Ronne), they have been combined. Then, points in the open ocean were added to the closest basin.

A mask of basin numbers is provided in:

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/imbie2/basin_numbers_ismip8km.nc

2.2.8 Buttressing flux response numbers

We provide Buttressing Flux Response Numbers and buttressing bins that are remapped for the ISMIP 8km grid. The bins are obtained by creating equally-sized bins of the BFRN that are larger than 0.02% (BFRN from Reese et al., 2018).

/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/meltmip/BFRN_ismip_8km.nc

2.2.9 ISMIP ocean grid

The grid as used in ISMIP7 for the oceanic forcing is similar to the ISIP6 grid, with 8km horizontal resolution and 60m vertical resolution. The grid is provided here:

```
/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/ismip/ismip_8km_60m_grid.nc
```

2.2.10 Topography

Most recent topography data, as used for the extrapolation of ocean conditions, is BedMap3 (<https://doi.org/10.5285/2d0e4791-8e20-46a3-80e4-f5f6716025d2>, Pritchard et al., 2025) and BedMachine version 3 (Morlighem, 2022). The files we use are remapped to the 8km ISMIP grid and provided here:

```
/ISMIP6/ISMIP7_Prep/AIS_ocean/share_with_modellers/topography/
```

```
BedMachineAntarctica-v3_ismip_8km.nc
```

```
BedMachineAntarctica-v3.nc
```

```
Bedmap3_ismip_8km.nc
```

```
bedmap3.nc
```

2.3 Calibration protocol for melt parameterisations

The idea of the calibration protocol is to help modellers to find three sets of parameters for their melt parameterisation that allow them to model 5th, mean, and 95th percentiles of melt rate evolution in the ISMIP projections. These parameter sets are calibrated with present-day melt rates as well as melt rates projected by dynamical ocean circulation models in case large-scale changes in ice shelf cavities occur, hence constraining parameters to provide reasonable melt rate sensitivities to oceanic changes. We provide examples of applying the protocol for the quadratic and PICO parameterisations, calibrating the thermal exchange parameter and the thermal exchange and overturning parameters, respectively. Please find the jupyter notebooks on github in the “meltmip” folder.

To ensure consistency with the datasets, modellers are asked to use a recent present day geometry (e.g., BedMap3 or BedMachine3) for the calibration rather than an ice sheet model initial state with substantially different grounding line position, ice thickness, or ice shelf extent. It is essential that the parameterisation is tuned and evaluated exactly as it is used in all the ISMIP7 ice sheet simulations, i.e. with the exact same routines and on the exact same grid.

For the calibration, modellers are required to run an ensemble of simulations with their melt parameterisation of choice for fixed bathymetry. In the ensemble, they vary the relevant parameters for their melt parameterisation within a plausible range. For example, for the quadratic melt parameterisation, this requires varying the K values (see section 2.1) while for PICO this requires varying both the overturning coefficient C and the heat exchange parameter γ_{AT} . If basin-wide ocean temperature input corrections are applied, as in Jourdain et al., 2020, we suggest pre-calculating those for each parameter (pair) using

observational present-day melt rates as otherwise the ensemble would grow out of proportion (see comment below). As a next step, the optimal parameters are determined by randomly sampling optimisation target values as specified in the following. Note that we are planning to develop a toolbox for the parameter sampling so that modellers do not have to develop their own scripts later on.

The parameters are obtained by minimising the following objective function

$$I = a_1 \frac{J_1}{\text{median}(J_1)} + a_2 \frac{J_2}{\text{median}(J_2)} + a_3 \frac{J_3}{\text{median}(J_3)},$$

where the terms J_i are detailed further below and the coefficients a_i are randomly sampled between 0 and 1 and should satisfy $a_1 + a_2 + a_3 = 1$ (This is achieved by randomly sampling the a_i between 0 and 1 from a uniform distribution, and then dividing each a_i by $a_1 + a_2 + a_3$). Random sampling of coefficients a_i varies the importance of each term J_i . This, together with further random sampling in the terms J_i , allows us to assess a range of parameters, from which the min, median, max are used to represent the 5th, 50th, 95th percentiles in the ISMIP experiments. We suggest sampling the a_1 , a_2 , a_3 and target melt rates in each term at least 10,000 times. Each term is normalised by the ensemble median $\text{median}(J_i)$.

The first term J_1 of the objective function assesses the fit with present-day, basin-aggregated melting using the mean absolute error

$$J_1 = \frac{1}{16} \sum_{b=1}^{16} |BMB_{\text{param}}(b) - BMB_{\text{obs}}(b)|,$$

where $BMB_{\text{param}}(b)$ is the modelled basal mass balance in Gt/a integrated over basin b and modelled using the present-day ocean climatology (see section 2.2.2), and $BMB_{\text{obs}}(b)$ is the observed basal mass balance in Gt/a integrated over basin b , sampled from a normal distribution with mean value and standard deviation from observations as described above (see section 2.2.3). Basins are the imbie2 basins as provided for ISMIP6 (see section 2.2.).

The second term J_2 of the objective function assess the fit with melting in buttressing relevant regions using the mean absolute error

$$J_2 = \frac{1}{10} \sum_{i=1}^{10} \frac{BFRN(i)}{\text{median}(BFRN)} |BMB_{\text{param}}(i) - BMB_{\text{obs}}(i)|,$$

where $BMB_{\text{param}}(i)$ and $BMB_{\text{obs}}(i)$ are the modelled and observed sub-shelf melt integrated over the i th buttressing bin, respectively (see section 2.2.7). Each bin is weighted with the median BFRN over that bin, $BFRN(i)$, relative to the median over all values $>0.02\%$ $\text{median}(BFRN)$. Observed melt rates are again sampled from a normal distribution with mean and standard deviation as detailed in section 2.2.3. This follows the binning approach of Menthon et al., 2025, with using buttressing bins instead of melt rate bins to make sure that melting in buttressing relevant regions is well represented.

The last term J_3 of the objective function assess the sensitivity of the melt in the parameterisation to ocean temperature changes as

$$J_3 = \frac{1}{2} \sum_{j=1}^n w_j (|BMB_{param,cold}(j) - BMB_{ref,cold}(j)| + |BMB_{param,warm}(j) - BMB_{ref,warm}(j)|)$$

where j is a randomly sampled representation of ocean modelling data, and $BMB_{param,cold}(j)$, $BMB_{param,warm}(j)$, $BMB_{ref,cold}(j)$, and $BMB_{ref,warm}(j)$ are the modelled (with the parameterisation) and reference (from ocean model) melt rates in Gt/a for a “warm” and a “cold” ocean state integrated over an imbie basin. The j ’s include each imbie2 basin from Mathiot et al. 2023; Timmermann & Göller 2017 for the Filchner-Ronne Ice Shelf,.. [further data to be added]. Reference data is sampled as for the other two terms from a normal distribution with mean given by the melt rate mean of each model, and standard deviation given by the uncertainty in the melt rates, calculated from the standard deviation of the modelled melt in the corresponding period normalised by the square root of the length of the period, i.e. $\frac{\sigma}{\sqrt{a}}$ with a being the number of years we calculate the standard deviation over (see section 2.2.4, 2.2.5). Usually, $a=20$. The weights w_j are randomly sampled from between 0 and 1, such that they sum to 1, similar to the a_i in term J_1 .

Although this is transparent for ice-sheet modellers, it should be noted that none of the temperature and salinity used in the J_i calculations are bias corrected. Regional ocean models do have biases, but we assume that melt rates are consistent with the ocean properties, i.e. underestimated if the ocean is cold biased as in Naughten et al. (2018). We are also aware that the ice shelf topography has varied in time and differs from Bedmap3/BedMachinev3 in the various data used to constrain the ice-shelf topography in the ocean simulations, but we consider that these are relatively small changes that are of second order compared to other uncertainties.

In the details, there are several ways to adjust the parameters, and modellers are welcome to adjust the objective function to their parameterisations. It should be noted that if temperature corrections δT_b are used in individual basins, they should be considered in a way or another in the parameter calibration. In ISMIP6, we chose to first tune the melt sensitivity to the thermal forcing, then we added corrections in every basin to match the present-day melt rates (Jourdain et al., 2020). The same approach is still possible in ISMIP7 but changing δT_b after the main parameters have been optimised using the objective function will impact the sensitivity of melt rates to warming (as it is after any correction of the thermal forcing). This can be avoided by using the opposite approach which consists of first identifying δT_b corrections needed to match observational present-day melt rates for any possible value of the (set of) parameter(s). Hence, the δT_b corrections are identified a number of times, and not just once, and vary between different parameter values of $K/\gamma T \& C$. After, the δT_b corrections are kept fixed for their corresponding parameter value/pair, and the optimisation suggested above is applied to identify the minimising parameter pairs where melt rates are determined using ocean forcing including the corresponding δT_b corrections. For example, for PICO we first optimise δT for any possible values of C and γT using the first term (J_1 , present-day, basin-wide melt), and then optimise all J_k terms together using an ensemble over C and γT . In any case, different values of the parameterisation coefficient(s) (e.g., different percentiles of K values in eq. 1) should have different δT_b corrections for a given basin.

Examples will be given for the quadratic melt parameterisation and pico in

<https://github.com/ismip/ismip7-antarctic-ocean-forcing.git>

`meltmip/parameter_selection_quadratic_example.ipynb`

`meltmip/parameter_selection_pico_example.ipynb`

3. MeltMIP - not required yet, included here for completeness

3.1 Motivation & aims

How to represent ocean-driven melting is one of the large uncertainties in ISMIP6 projections (Nowicki et al., 2020; Jourdain et al., 2020) and can for example explain differences between the ISMIP6 and LARMIP results (Seroussi et al., 2020; Levermann et al., 2020; Reese et al., 2021). Depending on the choice of parameterisation and parameters, projected melt rates can differ substantially (Jourdain et al., 2020, Seroussi et al., 2020, Burgard et al., 2022). How to best calibrate parameters in the melt parameterisation remains an open question (Burgard et al., 2022; Reese et al., 2023). Projected melt rates in ISMIP6 AIS 2300 (Seroussi et al., 2024), where the choice of melt parameterisation and parameters was left to the modellers, vary substantially in-between models for the same climate forcing.

Open questions relevant for ISMIP7 and related to the handling of melt parameterisations for ocean-driven forcing are:

- 1) How do the chosen parameterisations and parameters respond to changes in ocean forcing?
- 2) How does a chosen parameterisation with parameters respond to changes in ice shelf cavity geometry?
- 3) What influence does the ice sheet initial state have, i.e., how big is the difference between using BedMachine or similar dataset versus the native ice sheet initial state for calculating melt rates?
- 4) How is the behaviour of melt parameterizations affected by the ice sheet model resolution and grid?
- 5) How well does our suggested calibration protocol work, i.e., do parameterisations show similar melt sensitivities to ocean temperature increases?

We hence propose an Melt parameterisations Model Intercomparison Project (MeltMIP) with the following aims:

- (1) The first aim of MeltMIP is to provide a standard protocol for assessing the impact of a choice of sub-shelf melt parameterisation and parameters on melt rates and melt sensitivity.
- (2) The second aim is to better understand the diversity of ISMIP7 ice sheet projections, i.e. attribute a part of the diversity to the diversity of melt rates (either due to parameterisation choices, resolution or calibration).
- (3) The third aim is to have a physically-based parameter set for parameterisations that yields a plausible order of magnitude of melt rate changes in projections.

3.2 Experimental design

We suggest two tiers of experiments.

Tier 1: modellers follow our suggested parameter selection protocol for the melt parameterisation of their choice, see calibration design (section 2.3).

Tier 2: modellers use one set of their own choice of melt parameterisations and tuned parameters. Experiments with * do not apply to Tier 2.

Modellers are asked to minimally contribute to Tier 1. Modellers are welcome to submit more sets of experiments for each Tier with other parameters and/or parameterisations.

List of experiments:

In both Tiers, we have the following experiments:

Experiment	Description	Values	Purpose	#
BedMap_deltaT	Calculate melt rates with BedMap geometry and ocean climatology, adding a temperature perturbation deltaT	$\Delta T = 0, 0.5, 1, 2, 4$ deg C	How good is the calibration? 1) For deltaT=0 we have "present-day melt rates" that can be compared to observations, 2) From the deltaT>0 experiments, we can calculate the melt sensitivity	5
BedMap_deltaT_pc t*	Similar as above, but using melt parameters tuned for the 5th and 95th percentile	$\Delta T = 0, 0.5, 1, 2, 4$ deg C and pct=5, 95th	How good is the tuning and how do melt rates and melt sensitivities compare for all three sets of parameters?	10
BedMap_deltaT_resoX	Same as above, but with changing the ice sheet model grid resolution by a factor X.	$\Delta T = 0, 1$ deg C and X=0.5, 1, 2	Understand the robustness of melt rates and melt rate sensitivity in the melt parameterisation under changing ice sheet model resolution	6
Init_deltaT	Same as BedMap_deltaT but with the modellers choice of initial geometry	$\Delta T = 0, 1$ deg C	Understand whether ice geometry used has an influence on melt rates and melt sensitivity in the initialisation.	2
MISOMIP_deltaT	Same as BedMap_deltaT but with retreated MISOMIP geometry	$\Delta T = 0, 1$ deg C	Understand how parameterisations handle large-scale grounding line retreat and geometry changes.	2

Each experiment is a diagnostic solution of the melt parameterisation. No transient calculations are required. If the melt parameterisation has a time dependency, the equilibrium response should be used.

resoX: for a model resolution increasing by a factor X of the original one (i.e., for a structure grid, if your standard resolution is 16km, X=0.5 would mean you go to 8km; if you use an unstructured grid, either you refine all elements to be about half the size). We aim to use

X=0.5,1,2, the original (approximate) model resolution should be about the one that will be used in their ice sheet model projections and that modellers used for parameter calibration (basically this is the outcome of the optimal parameters).

Initi_deltaT: melt rates calculated on an ice sheet model initial state, e.g., the one for ISMIP7/histMIP/ISMIP6/the group's current preferred initial ice sheet state using the present-day climatology we provide (and the modellers \deltaT corrections if they use such), modellers use their native model resolution. Otherwise, similar to BedMap experiments.

Output format

Following the ISMIP7 standards, we ask for NetCDF files that contain 2 spatial dimensional variables of:

- sub-shelf melt rates,
- grounded/floating mask that indicates where melting is applied (and ice shelf extent),
- a mask with information on partially floating cells (if melting is applied in those)
- thermal forcing used at the ice shelf base to calculate melt rates (or ocean temperatures, salinities if using PICO, Plume etc).
- Ice geometry (thickness, draft depth, surface elevation)

Results should be regridded to the ISMIP grid, on 32km, 16km, 8km, 4km, 2km, 1km resolution depending on the model resolution.

We also ask modellers to fill in a standard sheet, with information about: model resolution, choice of parameterisation, parameters, deltaT (per basins), information on grounding zone melt, ..

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